



L^1 solution to scalar BSDEs with logarithmic sub-linear growth generators[☆]

Shengjun Fan^a, Ying Hu^{b,c}, Shanjian Tang^{d,*}

^a School of Mathematics, China University of Mining and Technology, Xuzhou 221116, China

^b Univ. Rennes, CNRS, IRMAR-UMR6625, F-35000, Rennes, France

^c School of Mathematical Sciences, Fudan University, Shanghai 200433, China

^d Department of Finance and Control Sciences, School of Mathematical Sciences, Fudan University, Shanghai 200433, China

ARTICLE INFO

Article history:

Received 9 July 2022

Received in revised form 1 May 2023

Accepted 6 May 2023

Available online xxxx

Keywords:

Backward stochastic differential equation

L^1 solution

Existence and uniqueness

Logarithmic sub-linear growth generator

Comparison theorem

ABSTRACT

With the test function method and a localization technique, a scalar backward stochastic differential equation (BSDE for short) subject to an L^1 terminal condition is shown to have an L^1 solution when the generator $g(t, y, z)$ has a one-sided linear growth in y and a logarithmic sub-linear growth in z , which improves some existing results. A new idea to study the existence of an adapted solution to a BSDE is given. When the generator $g(t, y, z)$ additionally has an extended monotonicity in y and a logarithmic uniform continuity in z , we further establish a comparison theorem for the L^1 solutions to the above BSDEs, which yields immediately the uniqueness of the solution.

© 2023 Published by Elsevier B.V.

1. Introduction

Fix a positive integer d and a real number $T > 0$. For $a, b \in \mathbb{R}$, define $a \wedge b := \min\{a, b\}$, $a^+ := \max\{a, 0\}$ and $a^- := -\min\{a, 0\}$, and $\text{sgn}(x) := \mathbf{1}_{x>0} - \mathbf{1}_{x\leq 0}$ with $\mathbf{1}_A$ being the indicator function of set A . Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a complete probability space with augmented filtration $(\mathcal{F}_t)_{t \in [0, T]}$ generated by a d -dimensional standard Brownian motion $(B_t)_{t \in [0, T]}$. For each $p > 0$, denote by $L^p(\mathcal{F}_t)$ for $t \in [0, T]$ the collection of all $\mathcal{F}_t$ -measurable real-valued random variables ξ satisfying $\mathbb{E}[|\xi|^p] < +\infty$ (in particular, $L^p(\mathcal{F}_T)$ is also denoted by L^p for simplicity), $\mathcal{L}^p$ the collection of all $(\mathcal{F}_t)$ -adapted real-valued processes $(X_t)_{t \in [0, T]}$ satisfying

$$\|X\|_{\mathcal{L}^p} := \left\{ \mathbb{E} \left[\left(\int_0^T |X_t| dt \right)^p \right] \right\}^{\frac{1}{p} \wedge 1} < +\infty,$$

[☆] Partially supported by National Natural Science Foundation of China (Nos. 12171471, 12031009 and 11631004), by Key Laboratory of Mathematics for Nonlinear Sciences (Fudan University), Ministry of Education, Handan Road 220, Shanghai 200433, China; by Lebesgue Center of Mathematics "Investissements d'avenir" program-ANR-11-LABX-0020-01, by CAESARS-ANR-15-CE05-0024 and by MFG-ANR-16-CE40-0015-01.

* Corresponding author.

E-mail addresses: f.s.j@126.com (S. Fan), ying.hu@univ-rennes1.fr (Y. Hu), sjtang@fudan.edu.cn (S. Tang).

$\mathcal{S}^p$ the collection of $(\mathcal{F}_t)$ -adapted continuous real-valued processes $(Y_t)_{t \in [0, T]}$ satisfying

$$\|Y\|_{\mathcal{S}^p} := \left(\mathbb{E} \left[\sup_{t \in [0, T]} |Y_t|^p \right] \right)^{\frac{1}{p} \wedge 1} < +\infty,$$

and $\mathcal{M}^p$ the collection of all $(\mathcal{F}_t)$ -adapted d -dimensional row vector processes $(Z_t)_{t \in [0, T]}$ satisfying

$$\|Z\|_{\mathcal{M}^p} := \left\{ \mathbb{E} \left[\left(\int_0^T |Z_t|^2 dt \right)^{p/2} \right] \right\}^{\frac{1}{p} \wedge 1} < +\infty.$$

Recall that an $(\mathcal{F}_t)$ -adapted real-valued process $(X_t)_{t \in [0, T]}$ belongs to class (D) if the $\{X_\tau : \tau \in \Sigma_T\}$ is a family of uniformly integrable random variables, where Σ_T represents the collection of all $(\mathcal{F}_t)$ -stopping times τ taking values in $[0, T]$. The equality and inequality between random elements are stood in the sense of $\mathbb{P}$ -a.s.

We study the following scalar backward stochastic differential equation (BSDE for short):

$$Y_t = \xi + \int_t^T g(s, Y_s, Z_s) ds - \int_t^T Z_s dB_s, \quad t \in [0, T], \quad (1.1)$$

where ξ is an $\mathcal{F}_T$ -measurable real-valued random variable, called the terminal condition, and the process $g(\cdot, \cdot, y, z) : \Omega \times [0, T] \rightarrow \mathbb{R}$ is $(\mathcal{F}_t)$ -adapted for each $(y, z) \in \mathbb{R} \times \mathbb{R}^{1 \times d}$ (the function g is called the generator). A pair of processes $(Y_t, Z_t)_{t \in [0, T]}$, taking values in $\mathbb{R} \times \mathbb{R}^{1 \times d}$, are called an adapted solution of (1.1), if

they are $(\mathcal{F}_t)$ -adapted such that $\mathbb{P}$ -a.s., $t \mapsto Y_t$ is continuous, $t \mapsto |g(t, Y_t, Z_t)| + |Z_t|^2$ is integrable, and (1.1) is satisfied.

Throughout the paper, we always suppose that $\beta \geq 0$ and $\gamma, c > 0$ are three given nonnegative constants, and $(f_t)_{t \in [0, T]}$ is a given $(\mathcal{F}_t)$ -adapted nonnegative process. By BSDE (ξ, g) , we mean the BSDE with terminal condition ξ and generator g . For convenience of exposition, let the generator g satisfy for $\alpha > 0$ the following growth condition:

$$|g(\omega, t, y, z)| \leq f_t(\omega) + \beta|y| + \gamma|z|^\alpha, \quad (\omega, t, y, z) \in \Omega \times [0, T] \times \mathbb{R} \times \mathbb{R}^{1 \times d}. \quad (1.2)$$

In each of the five cases $\alpha \in (0, 1)$, $\alpha = 1$, $\alpha \in (1, 2)$, $\alpha = 2$ and $\alpha > 2$, the generator g is usually called to have a linear growth in y and a sub-linear, linear, sub-quadratic, quadratic and super-quadratic, respectively, growth in z . The following existence results are well known. For the case of $\alpha \in (0, 1)$, it follows from [1,2] that if $(\xi, f) \in L^1 \times \mathcal{L}^1$, then BSDE (ξ, g) admits a solution $(Y_t, Z_t)_{t \in [0, T]}$ such that Y belongs to class (D) and $(Y, Z) \in \mathcal{S}^p \times \mathcal{M}^p$ for each $p \in (0, 1)$. For the case of $\alpha = 1$, if $(\xi, f) \in L^p \times \mathcal{L}^p$ for some $p > 1$, then BSDE (ξ, g) admits a solution $(Y, Z) \in \mathcal{S}^p \times \mathcal{M}^p$, see e.g. [1,3–6] for more details. For the case of $\alpha = 2$, by [7] we know that if (ξ, f) is bounded, then BSDE (ξ, g) admits a solution $(Y_t, Z_t)_{t \in [0, T]}$ such that Y is a bounded process and $Z \in \mathcal{M}^2$. For the case of $\alpha > 2$, it follows from [8] that even if (ξ, f) is bounded, BSDE (ξ, g) may not admit a solution.

When the generator g satisfies (1.2) with $\alpha = 2$, [9] gave a general existence result on an adapted solution to BSDE (ξ, g) , which weakens the boundedness condition on (ξ, f) in [7] to a certain exponential moment condition. To the best of our knowledge, this is the first work investigating the weakest possible integrability condition on (ξ, f) for existence of an adapted solution to BSDE (ξ, g) when the generator g has a certain growth condition on both unknown variables. In this spirit, [10,11] studied the same problem in the case that the generator g satisfies (1.2) with $\alpha = 1$ by applying the dual representation of solution to BSDE with convex generator used in [4,12] and utilizing the test function method respectively. Very recently, the case that the generator g satisfies (1.2) with $\alpha \in (1, 2)$ was tracked in [13], and the case that the generator g satisfies (1.2) with $|z|^\alpha$ instead of $|z|^\alpha$ was dealt with in [14], where g has a super-linear growth in z . We would like to especially mention that in some of known results mentioned above, the generator g can be allowed to have a one-sided linear growth in y .

Generally speaking, more assumptions are required so as to obtain the uniqueness and the comparison theorem of solutions of scalar BSDEs. In particular, when the generator $g(t, y, z)$ has a linear growth in z , an extended monotonicity in y and the uniform continuity in z are additionally supposed to obtain the desired results (see [2,4,10,15,16] for details, where Gronwall's inequality, Bihari's inequality and Girsanov's transform are usually used as important tools). And, when the generator g has a super-linear, sub-quadratic or quadratic growth in z , some extended convexity (concavity) of the generator g in (y, z) are additionally assumed to get the uniqueness or the comparison theorem (see [2,13,14,17–19] for more details, where the θ difference method and the a priori estimate technique usually play a key role in the proof). In addition, we would like to mention that under the growth condition (1.2) with $\alpha \in (0, 1)$, if the generator $g(t, y, z)$ further has an extended monotonicity in y and the uniform continuity in z , then the uniqueness and the comparison theorem hold true for solutions of scalar BSDEs, see [1,2] for details.

In this paper, we put forward a new idea to study the existence of an adapted solution to BSDE (ξ, g) . More specially, the following question is the starting point of this paper. That is, if only $(\xi, f) \in L^1 \times \mathcal{L}^1$ is satisfied, then what growth conditions (as weakest as possible) of the generator g in (y, z) can ensure existence of the

adapted solution to BSDE (ξ, g) ? As mentioned before, [1] gave an answer of this problem where the generator g has a linear growth in y and a sub-linear growth in z , i.e., g satisfies (1.2) with $\alpha \in (0, 1)$. Then, a question naturally arises: can we find some weaker conditions? This paper gives an affirmative answer of this question. Roughly speaking, by utilizing the localization technique put forward initially in [9], we prove that BSDE (ξ, g) admits a solution $(Y_t, Z_t)_{t \in [0, T]}$ such that Y belongs to class (D) and $(Y, Z) \in \mathcal{S}^p \times \mathcal{M}^p$ for each $p \in (0, 1)$ when $(\xi, f) \in L^1 \times \mathcal{L}^1$ and the generator g has a one-sided linear growth in y and a logarithmic sub-linear growth in z , see Theorem 2.1 in Section 2. It is crucial in the proof to search for a suitable test function (see Proposition 3.3 in Section 3) to apply Itô's formula and then obtain the desired a priori bound on the first parts of adapted solutions to the approached BSDEs. For this purpose, we appeal an important technical inequality, see Proposition 3.2 in Section 3. Furthermore, by using the same a priori estimate technique as above and combining Theorem 2.1 in [2], we establish a comparison theorem for the adapted L^1 solutions to the previous BSDEs when the generator g additionally satisfies an extended monotonicity condition in y and a logarithmic uniform continuity condition in z , which naturally yields the uniqueness of the L^1 solution, see Proposition 2.5 in Section 2.

The rest of this paper is organized as follows. In the next section we state the main result, and in Section 3 we give the proof.

2. Statement of the main result

Let us first introduce the following assumptions on the generator g .

- (H1) $d\mathbb{P} \times dt - a.e.$, $g(\omega, t, \cdot, \cdot)$ is continuous.
- (H2) g has a one-sided linear growth in y and a logarithmic sub-linear growth in z , i.e., there exists a constant $\lambda \in (1/2, 1]$ that $d\mathbb{P} \times dt - a.e.$, for each $(y, z) \in \mathbb{R} \times \mathbb{R}^{1 \times d}$,
$$\text{sgn}(y)g(\omega, t, y, z) \leq f_t(\omega) + \beta|y| + \frac{\gamma|z|}{(\ln(e + |z|))^\lambda}.$$
- (H3) g has a general growth in (y, z) , i.e., there exists a continuous nondecreasing function $h(\cdot) : [0, +\infty) \rightarrow [0, +\infty)$ with $h(0) = 0$ such that $d\mathbb{P} \times dt - a.e.$, for each $(y, z) \in \mathbb{R} \times \mathbb{R}^{1 \times d}$,
$$|g(\omega, t, y, z)| \leq f_t(\omega) + h(|y|) + c|z|^2.$$
- (H4) g satisfies an extended monotonicity condition in y , i.e., there exists a continuous, nondecreasing and concave function $\rho(\cdot) : [0, +\infty) \rightarrow [0, +\infty)$ with $\rho(0) = 0$, $\rho(u) > 0$ for $u > 0$, and $\int_{0+} \frac{du}{\rho(u)} = +\infty$ such that $d\mathbb{P} \times dt - a.e.$, for each $(y_1, y_2, z) \in \mathbb{R} \times \mathbb{R} \times \mathbb{R}^{1 \times d}$,
$$\text{sgn}(y_1 - y_2)(g(\omega, t, y_1, z) - g(\omega, t, y_2, z)) \leq \rho(|y_1 - y_2|).$$
- (H5) g satisfies a logarithmic uniform continuity condition in z , i.e., there exists a nondecreasing continuous function $\kappa(\cdot) : [0, +\infty) \rightarrow [0, +\infty)$ with linear growth and $\kappa(0) = 0$ such that $d\mathbb{P} \times dt - a.e.$, for each $(y, z_1, z_2) \in \mathbb{R} \times \mathbb{R}^{1 \times d} \times \mathbb{R}^{1 \times d}$,
$$|g(\omega, t, y, z_1) - g(\omega, t, y, z_2)| \leq \kappa\left(\frac{|z_1 - z_2|}{(\ln(e + |z_1 - z_2|))^\lambda}\right),$$
where the constant $\lambda \in (1/2, 1]$ is the same as in (H2).

The following existence and uniqueness theorem is the main result of this paper.

Theorem 2.1. *Let the terminal condition $\xi \in L^1$ and the generator g satisfy (H1)–(H3) with $f \in \mathcal{L}^1$. Then BSDE (ξ, g) admits a solution*

$(Y_t, Z_t)_{t \in [0, T]}$ such that Y belongs to class (D) and $(Y, Z) \in \mathcal{S}^p \times \mathcal{M}^p$ for each $p \in (0, 1)$. And, there exists a constant $C > 0$ depending only on $(\beta, \gamma, \lambda, T)$ such that

$$|Y_t| \leq |Y_0| + \int_0^t f_s ds \leq C \mathbb{E} \left[|\xi| + \int_0^T f_t dt \middle| \mathcal{F}_t \right] + C, \quad t \in [0, T]. \quad (2.1)$$

Moreover, if g also satisfies assumptions (H4) and (H5), then the solution (Y, Z) such that Y belongs to class (D) is unique.

Remark 2.2. With respect to Theorem 2.1, we make the following several remarks.

- (i) It is not hard to verify that for each $\alpha \in (0, 1)$, there exists a constant $k_\alpha > 0$ depending only on α such that for each $z \in \mathbb{R}^{1 \times d}$,

$$|z|^\alpha \leq k_\alpha + \frac{|z|}{\ln(e + |z|)}.$$

This means that for the one-dimensional case, (H2) is weaker than the sub-linear growth assumption (H6) of [1], and then Theorem 2.1 improves the related existence result of [1].

- (ii) Note that for each $\lambda > 1$ and $z \in \mathbb{R}^{1 \times d}$,

$$\frac{|z|}{(\ln(e + |z|))^\lambda} \leq \frac{|z|}{\ln(e + |z|)}.$$

Therefore, the constant λ appearing in (H2) can take values in $(1/2, +\infty)$. However, it is still open whether Theorem 2.1 is still true when $\lambda \in (0, 1/2]$.

- (iii) It is not hard to verify that the constant e appearing in (H2) and (H5) can be replaced with any constant bigger than 1. And, it is clear that if g satisfies (H5), then it is uniformly continuous in z . In addition, it can be proved that (H3)–(H5) imply (H2).

In the sequel, we give two examples to which Theorem 2.1 but no existing result applies.

Example 2.3. For each $(\omega, t, y, z) \in \Omega \times [0, T] \times \mathbb{R} \times \mathbb{R}^{1 \times d}$, define

$$g(\omega, t, y, z) := e^{B_t(\omega)} + e^{-y} \cos^2 |z| + \frac{|z| \sin |z|}{(\ln(e + |z|))^{2/3}} - |z|^2 \sin y.$$

It is easy to check that g satisfies (H1)–(H3) with

$$f = e^B + 1, \quad \beta = 0, \quad \gamma = 1, \quad \lambda = 2/3, \quad c = 1 \quad \text{and} \quad h(u) = e^u.$$

It then follows from Theorem 2.1 that for each $\xi \in L^1$, BSDE(ξ, g) admits an adapted solution $(Y_t, Z_t)_{t \in [0, T]}$ such that Y belongs to class (D) and $(Y, Z) \in \mathcal{S}^p \times \mathcal{M}^p$ for each $p \in (0, 1)$.

Example 2.4. For each $(\omega, t, y, z) \in \Omega \times [0, T] \times \mathbb{R} \times \mathbb{R}^{1 \times d}$, define

$$\bar{g}(\omega, t, y, z) := B_t(\omega) + y^2 \mathbf{1}_{y \leq 0} + l(|y|) + l \left(\frac{|z|}{\ln(e + |z|)} \right) + \sqrt{|z|},$$

where

$$l(u) := u |\ln u| \mathbf{1}_{0 \leq u \leq \varepsilon} + l'(\varepsilon)(u - l(\varepsilon)) \mathbf{1}_{u > \varepsilon}, \quad u \in (0, +\infty)$$

with $\varepsilon > 0$ being sufficiently small. Note that if $F(\cdot) : [0, +\infty) \rightarrow [0, +\infty)$ is a nondecreasing concave function with $F(0) = 0$, then for each $0 < x < y$, by letting $k := x/y \in (0, 1)$ we have

$$\begin{aligned} F(y) - F(x) &= F(y) - F(ky + (1 - k) \cdot 0) \\ &\leq F(y) - (kF(y) + (1 - k)F(0)) \\ &= (1 - k)F(y) + kF(0) \\ &\leq F((1 - k)y + k \cdot 0) = F(y - x), \end{aligned}$$

and then for each $x_1, x_2 \geq 0$,

$$|F(x_1) - F(x_2)| \leq F(|x_1 - x_2|).$$

With the above observation, it is not very hard to verify that for each $x_1, x_2 \geq 0$,

$$|l(x_1) - l(x_2)| \leq l(|x_1 - x_2|) \quad \text{and} \quad \int_{0+} \frac{du}{l(u)} = +\infty,$$

$$\left| \frac{x_1}{\ln(e + x_1)} - \frac{x_2}{\ln(e + x_2)} \right| \leq \frac{|x_1 - x_2|}{\ln(e + |x_1 - x_2|)}$$

and

$$\begin{aligned} |\sqrt{x_1} - \sqrt{x_2}| &\leq \sqrt{|x_1 - x_2|} \\ &\leq 2 \sqrt{\frac{|x_1 - x_2|}{\ln(e + |x_1 - x_2|)}} + \frac{|x_1 - x_2|}{\ln(e + |x_1 - x_2|)}. \end{aligned}$$

Thus, it can be checked that this $\bar{g}$ satisfies (H1)–(H5) with

$$f = B + 2, \quad \beta = 1, \quad \gamma = 2, \quad \lambda = 1, \quad c = 2, \quad h(u) = u^2 + u, \\ \rho(u) = l(u) \quad \text{and} \quad \kappa(u) = l(u) + u + 2\sqrt{u}.$$

It then follows from Theorem 2.1 that for each $\xi \in L^1$, BSDE(ξ, g) admits a unique adapted solution $(Y_t, Z_t)_{t \in [0, T]}$ such that Y belongs to class (D) and $(Y, Z) \in \mathcal{S}^p \times \mathcal{M}^p$ for each $p \in (0, 1)$.

The following proposition establishes a comparison result for the L^1 solutions of two BSDEs, which naturally yields the uniqueness part of Theorem 2.1.

Proposition 2.5. Let ξ and ξ' be two terminal conditions, g and g' be two generators, and $(Y_t, Z_t)_{t \in [0, T]}$ and $(Y'_t, Z'_t)_{t \in [0, T]}$ be respectively an adapted solution to BSDE(ξ, g) and BSDE(ξ', g'). Assume that g (resp. g') satisfies (H4) and (H5), and $(Y - Y')^+$ belongs to class (D). If $\xi \leq \xi'$ and

$$\begin{aligned} \mathbf{1}_{Y_t > Y'_t} (g(t, Y'_t, Z'_t) - g'(t, Y'_t, Z'_t)) &\leq 0 \\ \text{(resp. } \mathbf{1}_{Y_t > Y'_t} (g(t, Y_t, Z_t) - g'(t, Y_t, Z_t)) &\leq 0), \end{aligned} \quad (2.2)$$

then for each $t \in [0, T]$, we have $Y_t \leq Y'_t$.

We finish the section with the following application of Theorem 2.1 and Proposition 2.5: the notion of (conditional) g -expectation of [20] defined on the space $L^2(\mathcal{F}_T)$ of squarely integrable random variables, is extended to the larger one $L^1(\mathcal{F}_T)$ of integrable random variables.

Definition 2.6. Let the generator g satisfy assumptions (H1)–(H5) with $f \in \mathcal{L}^1$. Suppose further that g satisfies the following assumption:

$$d\mathbb{P} \times dt - a.e., \quad g(\omega, t, y, 0) \equiv 0, \quad \forall y \in \mathbb{R}. \quad (2.3)$$

By virtue of Theorem 2.1, for each $\xi \in L^1(\mathcal{F}_T)$ we can denote

$$\mathcal{E}_g[\xi | \mathcal{F}_t] := Y_t^\xi, \quad t \in [0, T], \quad (2.4)$$

where $(Y_t^\xi, Z_t^\xi)_{t \in [0, T]}$ is the unique solution of BSDE(ξ, g) such that Y^ξ belongs to class (D) and $(Y^\xi, Z^\xi) \in \mathcal{S}^p \times \mathcal{M}^p$ for each $p \in (0, 1)$. We call $\mathcal{E}_g[\xi | \mathcal{F}_t]$ the conditional g -expectation of ξ with respect to $\mathcal{F}_t$, and $\mathcal{E}_g[\xi] := \mathcal{E}_g[\xi | \mathcal{F}_0]$ the g -expectation of ξ .

Clearly, for each $t \in [0, T]$, the conditional g -expectation operator $\mathcal{E}_g[\cdot | \mathcal{F}_t]$ defined by (2.4) maps $L^1(\mathcal{F}_T)$ to $L^1(\mathcal{F}_t)$, which shares the same domain with the classical mathematical expectation operator. Furthermore, proceeding identically as in [21,22], we easily (thus omitting the proof) have from Theorem 2.1 and Proposition 2.5 together with (2.3) the following two propositions.

Proposition 2.7. $\mathcal{E}_g[\cdot]$ has the following properties:

- (i) (preserving of constants): For each constant $c \in \mathbb{R}$, $\mathcal{E}_g[c] = c$;
- (ii) (monotonicity): For each $\xi_1, \xi_2 \in L^1(\mathcal{F}_T)$, if $\xi_1 \geq \xi_2$, then $\mathcal{E}_g[\xi_1] \geq \mathcal{E}_g[\xi_2]$.

Proposition 2.8. For each $t \in [0, T]$, $\mathcal{E}_g[\cdot|\mathcal{F}_t]$ has the following properties:

- (i) If $\xi \in L^1(\mathcal{F}_t)$, then $\mathcal{E}_g[\xi|\mathcal{F}_t] = \xi$;
- (ii) For each $\xi_1, \xi_2 \in L^1(\mathcal{F}_t)$, if $\xi_1 \geq \xi_2$, then $\mathcal{E}_g[\xi_1|\mathcal{F}_t] \geq \mathcal{E}_g[\xi_2|\mathcal{F}_t]$;
- (iii) For each $\xi \in L^1(\mathcal{F}_T)$ and $r \in [0, T]$, we have $\mathcal{E}_g[\mathcal{E}_g[\xi|\mathcal{F}_t]|\mathcal{F}_r] = \mathcal{E}_g[\xi|\mathcal{F}_{t \wedge r}]$;
- (iv) For each $A \in \mathcal{F}_t$ and $\xi \in L^1(\mathcal{F}_T)$, $\mathcal{E}_g[\mathbf{1}_A \xi|\mathcal{F}_t] = \mathbf{1}_A \mathcal{E}_g[\xi|\mathcal{F}_t]$ and $\mathcal{E}_g[\mathbf{1}_A \xi] = \mathcal{E}_g[\mathbf{1}_A \mathcal{E}_g[\xi|\mathcal{F}_t]]$.

Both propositions indicate that the (conditional) g -expectation defined by (2.4) preserves essential properties (but except linearity) of the classical mathematical expectation. Further extensive issues on the (conditional) g -expectation still remain to be studied along the lines of [20–22].

3. Proof of the main result

We first introduce the following technical lemma.

Lemma 3.1. For each $x \geq 0$, we have

$$(e+x)\ln(e+x) > 2x \quad (3.1)$$

and

$$(e+x)^2 - 2x\ln(e+x) > e. \quad (3.2)$$

Proof. Let $f(x) := (e+x)\ln(e+x) - 2x$, $x \in [0, +\infty)$. A simple calculate gives

$$f'(x) = \ln(e+x) - 1 \geq 0, \quad x \geq 0.$$

Then, (3.1) follows since $f(x) \geq f(0) = e > 0$ for each $x \geq 0$.

In order to prove (3.2), we let $h(x) := (e+x)^2 - 2x\ln(e+x) - e$, $x \in [0, +\infty)$. A simple calculate gives that for each $x \geq 0$,

$$h'(x) = 2(e+x) - 2\ln(e+x) - \frac{2x}{e+x}$$

and

$$h''(x) = 2 - \frac{2}{e+x} - \frac{2e}{(e+x)^2} \geq 2 - \frac{4}{e} > 0.$$

Hence, for each $x \geq 0$ we have

$$h'(x) \geq h'(0) = 2e - 2 > 0$$

and then

$$h(x) \geq h(0) = e^2 - e > 0,$$

which yields the desired inequality (3.2). $\square$

By virtue of the last lemma we can prove the following key inequality.

Proposition 3.2. For each $x, y \geq 0$ and $\lambda \in [0, 1]$, it holds that

$$\frac{xy}{(\ln(e+y))^\lambda} \leq \frac{2x^2}{(\ln(e+x))^{2\lambda}} + \frac{1}{2}y^2. \quad (3.3)$$

Proof. The conclusion is obvious in the case of $\lambda = 0$. Let us prove the case of $\lambda \in (0, 1]$. For $(x, y) \in [0, +\infty) \times [0, +\infty)$, define the

function

$$\begin{aligned} f(x, y) &:= y^2 - \frac{2xy}{(\ln(e+y))^\lambda} + \frac{4x^2}{(\ln(e+x))^{2\lambda}} \\ &= \left(y - \frac{x}{(\ln(e+y))^\lambda}\right)^2 + \frac{4x^2}{(\ln(e+x))^{2\lambda}} - \frac{x^2}{(\ln(e+y))^{2\lambda}}. \end{aligned} \quad (3.4)$$

Clearly, in order to prove (3.3), it suffices to prove that $f(x, y) \geq 0$ for each $x, y \geq 0$. By (3.4) it is obvious when $y \geq x$, hence we only need to prove that $f(x, y) \geq 0$ for each $x \in (0, +\infty)$ and $y \in [0, x]$.

Now, fix arbitrarily $x \in (0, +\infty)$ and let $\bar{f}(y) := f(x, y)$, $y \in [0, x]$. Then we have

$$\bar{f}'(y) = 2y - \frac{2x}{(\ln(e+y))^\lambda} + \frac{2\lambda xy}{(e+y)(\ln(e+y))^{\lambda+1}}, \quad y \in [0, x] \quad (3.5)$$

and in view of (3.1),

$$\begin{aligned} \bar{f}''(y) &= 2 + \frac{2\lambda x}{(e+y)(\ln(e+y))^{\lambda+1}} + \frac{2\lambda x(e\ln(e+y) - (\lambda+1)y)}{(e+y)^2(\ln(e+y))^{\lambda+2}} \\ &= 2 + \frac{2\lambda x[(e+y)\ln(e+y) + e\ln(e+y) - (\lambda+1)y]}{(e+y)^2(\ln(e+y))^{\lambda+2}} \\ &> 2 + \frac{2\lambda x[(e+y)\ln(e+y) - 2y]}{(e+y)^2(\ln(e+y))^{\lambda+2}} > 0, \quad y \in [0, x]. \end{aligned} \quad (3.6)$$

The previous inequality (3.6) means that $\bar{f}$ is a strictly convex function on $[0, x]$. Furthermore, noticing by (3.5) that $\bar{f}'(0) = -2x < 0$ and

$$\bar{f}'(x) = 2x - \frac{2x}{(\ln(e+x))^\lambda} + \frac{2\lambda x^2}{(e+x)(\ln(e+x))^{\lambda+1}} > 0,$$

we deduce that there exists a unique $y_0 \in (0, x)$ such that $\bar{f}'(y_0) = 0$ and

$$f(x, y) = \bar{f}(y) \geq \bar{f}(y_0) = f(x, y_0), \quad y \in [0, x]. \quad (3.7)$$

In the sequel, let

$$y_1 = \frac{x}{2(\ln(e+y_1))^\lambda}. \quad (3.8)$$

Then, $y_1 \in (0, x/2)$ and it follows from (3.5), (3.1) and (3.8) that

$$\begin{aligned} \bar{f}'(y_1) &= \frac{x}{(\ln(e+y_1))^\lambda} - \frac{2x}{(\ln(e+y_1))^\lambda} + \frac{2\lambda xy_1}{(e+y_1)(\ln(e+y_1))^{\lambda+1}} \\ &= \frac{x[2\lambda y_1 - (e+y_1)\ln(e+y_1)]}{(e+y_1)(\ln(e+y_1))^{\lambda+1}} \\ &\leq \frac{x[2y_1 - (e+y_1)\ln(e+y_1)]}{(e+y_1)(\ln(e+y_1))^{\lambda+1}} < 0. \end{aligned}$$

Therefore, $y_1 < y_0$ and then it follows from (3.4) and (3.7) that

$$\begin{aligned} f(x, y) &\geq f(x, y_0) \\ &= \left(y_0 - \frac{x}{(\ln(e+y_0))^\lambda}\right)^2 + \frac{4x^2}{(\ln(e+x))^{2\lambda}} - \frac{x^2}{(\ln(e+y_0))^{2\lambda}} \\ &\geq \frac{4x^2}{(\ln(e+x))^{2\lambda}} - \frac{x^2}{(\ln(e+y_1))^{2\lambda}}, \quad y \in [0, x]. \end{aligned} \quad (3.9)$$

Finally, in view of $\lambda \in (0, 1]$ and $y_1 > 0$, by (3.2) we can deduce that

$$(e+y_1)^{2\frac{1}{\lambda}} - 2y_1(\ln(e+y_1))^\lambda \geq (e+y_1)^2 - 2y_1\ln(e+y_1) > e.$$

It follows from the last inequality and (3.8) that

$$2^{\frac{1}{\lambda}} \ln(e + y_1) \geq \ln(e + 2y_1(\ln(e + y_1))^\lambda) = \ln(e + x),$$

which means that

$$4(\ln(e + y_1))^{2\lambda} \geq (\ln(e + x))^{2\lambda}. \quad (3.10)$$

Combining (3.9) and (3.10) yields that $f(x, y) \geq 0$ for each $x \in (0, +\infty)$ and $y \in [0, x]$, which is the desired assertion. $\square$

Now, we fix arbitrarily $\lambda \in (1/2, 1]$ and search for a positive, continuous, strictly increasing and strictly convex function $\phi(s, x) : [0, T] \times [0, +\infty) \rightarrow (0, +\infty)$ satisfying

$$-\beta\phi_x(s, x)x - \phi_x(s, x)\frac{\gamma|z|}{(\ln(e + |z|))^\lambda} + \frac{1}{2}\phi_{xx}(s, x)|z|^2 + \phi_s(s, x) \geq 0, \quad (s, x, z) \in [0, T] \times [0, +\infty) \times \mathbb{R}^{1 \times d}, \quad (3.11)$$

where and hereafter, $\phi_s(\cdot, \cdot)$ denotes the first-order partial derivative of $\phi(\cdot, \cdot)$ with respect to the first variable, and $\phi_x(\cdot, \cdot)$ and $\phi_{xx}(\cdot, \cdot)$ respectively the first-order and second-order partial derivative of $\phi(\cdot, \cdot)$ with respect to the second variable. Observe from (3.3) that

$$\begin{aligned} & -\phi_x(s, x)\frac{\gamma|z|}{(\ln(e + |z|))^\lambda} + \frac{1}{2}\phi_{xx}(s, x)|z|^2 \\ &= \phi_{xx}(s, x)\left(-\frac{\gamma\phi_x(s, x)}{\phi_{xx}(s, x)}\frac{|z|}{(\ln(e + |z|))^\lambda} + \frac{1}{2}|z|^2\right) \\ &\geq -2\gamma^2\frac{\phi_x^2(s, x)}{\phi_{xx}(s, x)}\left(\ln\left(e + \frac{\gamma\phi_x(s, x)}{\phi_{xx}(s, x)}\right)\right)^{-2\lambda}. \end{aligned}$$

Hence, it is sufficient for the function $\phi(\cdot, \cdot)$ to satisfy that for each $(s, x) \in [0, T] \times [0, +\infty)$,

$$-\beta\phi_x(s, x)x - 2\gamma^2\frac{\phi_x^2(s, x)}{\phi_{xx}(s, x)}\left(\ln\left(e + \frac{\gamma\phi_x(s, x)}{\phi_{xx}(s, x)}\right)\right)^{-2\lambda} + \phi_s(s, x) \geq 0. \quad (3.12)$$

In the sequel, we choose the following test function

$$\phi(s, x) := (k + x)\left(1 - \frac{1}{(\ln(k + x))^\delta}\right)\mu_s, \quad (s, x) \in [0, T] \times [0, +\infty)$$

to explicitly solve the inequality (3.12), where δ is a positive constant to be assigned, $\mu_s : [0, T] \rightarrow (0, +\infty)$ is a nondecreasing and continuous differentiable function to be assigned and

$$k := \exp\left(\frac{2}{\delta} + 2(\delta + 1) + \left(\frac{2\delta}{\gamma}\right)^{\frac{1}{\delta+1}}\right). \quad (3.13)$$

First of all, a simple computation gives that for each $(s, x) \in [0, T] \times [0, +\infty)$,

$$\phi_x(s, x) = \left[1 - \frac{1}{(\ln(k + x))^\delta}\left(1 - \frac{\delta}{\ln(k + x)}\right)\right]\mu_s > 0,$$

$$\phi_{xx}(s, x) = \frac{\delta}{(k + x)(\ln(k + x))^{\delta+1}}\left(1 - \frac{\delta + 1}{\ln(k + x)}\right)\mu_s > 0,$$

and

$$\phi_s(s, x) = (k + x)\left(1 - \frac{1}{(\ln(k + x))^\delta}\right)\mu'_s > 0.$$

Combining (3.13) and the last three inequalities yields that for $(s, x) \in [0, T] \times [0, +\infty)$,

$$\frac{1}{2}\mu_s \leq \phi_x(s, x) \leq \mu_s, \quad (3.14)$$

$$\frac{\delta\mu_s}{2(k + x)(\ln(k + x))^{\delta+1}} \leq \phi_{xx}(s, x) \leq \frac{\delta\mu_s}{(k + x)(\ln(k + x))^{\delta+1}} \quad \text{and} \quad (3.15)$$

$$\phi_s(s, x) \geq \frac{1}{2}(k + x)\mu'_s. \quad (3.16)$$

It follows from (3.14), (3.13) and (3.15) that

$$\frac{\gamma\phi_x(s, x)}{\phi_{xx}(s, x)} \geq \frac{\gamma}{2\delta}(k + x)(\ln(k + x))^{\delta+1} \geq k + x. \quad (3.17)$$

Substituting (3.14)–(3.17) into the left hand side of (3.12) we can deduce that

$$\begin{aligned} & -\beta\phi_x(s, x)x - 2\gamma^2\frac{\phi_x^2(s, x)}{\phi_{xx}(s, x)}\left(\ln\left(e + \frac{\gamma\phi_x(s, x)}{\phi_{xx}(s, x)}\right)\right)^{-2\lambda} + \phi_s(s, x) \\ &\geq -\beta(k + x)\mu_s - \frac{2\gamma^2\mu_s^2}{\frac{\delta\mu_s}{2(k + x)(\ln(k + x))^{\delta+1}}(\ln(e + k + x))^{2\lambda}} + \frac{1}{2}(k + x)\mu'_s \\ &\geq (k + x)\left[-\left(\beta + \frac{4\gamma^2}{\delta}(\ln(k + x))^{\delta-(2\lambda-1)}\right)\mu_s + \frac{1}{2}\mu'_s\right], \\ &\quad (s, x) \in [0, T] \times [0, +\infty). \end{aligned}$$

Thus, if we pick $\delta = 2\lambda - 1$ and

$$\begin{aligned} \mu_s &:= \exp\left[2\left(\beta + \frac{4\gamma^2}{\delta}\right)s\right] \\ &= \exp\left[2\left(\beta + \frac{4\gamma^2}{2\lambda - 1}\right)s\right], \quad s \in [0, T], \end{aligned}$$

then (3.12) and then (3.11) holds.

In conclusion, we have the following proposition on the test function.

Proposition 3.3. *Let $\lambda \in (1/2, 1]$ and*

$$k_0 := \exp\left(\frac{2}{2\lambda - 1} + 4\lambda + \left(\frac{4\lambda - 2}{\gamma}\right)^{\frac{1}{2\lambda}}\right). \quad (3.18)$$

For $(s, x) \in [0, T] \times [0, +\infty)$, define

$$\begin{aligned} \varphi(s, x) &:= (k_0 + x)\left(1 - \frac{1}{(\ln(k_0 + x))^{2\lambda-1}}\right)\exp\left[2\left(\beta + \frac{4\gamma^2}{2\lambda - 1}\right)s\right]. \end{aligned} \quad (3.19)$$

Then, the test function $\varphi(\cdot, \cdot)$ satisfies that for each $(s, x, z) \in [0, T] \times [0, +\infty) \times \mathbb{R}^{1 \times d}$,

$$-\beta\varphi_x(s, x)x - \varphi_x(s, x)\frac{\gamma|z|}{(\ln(e + |z|))^\lambda} + \frac{1}{2}\varphi_{xx}(s, x)|z|^2 + \varphi_s(s, x) \geq 0. \quad (3.20)$$

The following Proposition 3.4 establishes an important a priori estimate for the solution to a BSDE.

Proposition 3.4. *Assume that ξ is a terminal condition, g is a generator satisfying (H2), and $(Y_t, Z_t)_{t \in [0, T]}$ is a solution of BSDE(ξ, g). If the process $|Y_t| + \int_0^t f_s ds$ belongs to class (D), then there exists a constant $C > 0$ depending only on $(\beta, \gamma, \lambda, T)$ such that for each $t \in [0, T]$,*

$$|Y_t| \leq |Y_t| + \int_0^t f_s ds \leq C\mathbb{E}\left[|\xi| + \int_0^T f_t dt \middle| \mathcal{F}_t\right] + C. \quad (3.21)$$

Proof. Define

$$\bar{Y}_t := |Y_t| + \int_0^t f_s ds \quad \text{and} \quad \bar{Z}_t := Z_t \text{sgn}(Y_t), \quad t \in [0, T].$$

Itô–Tanaka’s formula yields

$$\begin{aligned} \bar{Y}_t &= \bar{Y}_T + \int_t^T (\text{sgn}(Y_s)g(s, Y_s, Z_s) - f_s) ds \\ &\quad - \int_t^T \bar{Z}_s dB_s - \int_t^T dL_s, \quad t \in [0, T], \end{aligned}$$

where L is the local time of Y at 0. Now, let the function $\varphi(\cdot, \cdot)$ be defined in (3.19) and apply Itô–Tanaka’s formula to the process $\varphi(s, \bar{Y}_s)$ to deduce, in view of (H2),

$$\begin{aligned} d\varphi(s, \bar{Y}_s) &= \varphi_x(s, \bar{Y}_s) (-\text{sgn}(Y_s)g(s, Y_s, Z_s) + f_s) ds + \varphi_x(s, \bar{Y}_s) \bar{Z}_s dB_s \\ &\quad + \varphi_x(s, \bar{Y}_s) dL_s + \frac{1}{2} \varphi_{xx}(s, \bar{Y}_s) |\bar{Z}_s|^2 ds + \varphi_s(s, \bar{Y}_s) ds \\ &\geq \left[-\varphi_x(s, \bar{Y}_s) \left(\beta |Y_s| + \frac{\gamma |Z_s|}{(\ln(e + |Z_s|))^\lambda} \right) \right. \\ &\quad \left. + \frac{1}{2} \varphi_{xx}(s, \bar{Y}_s) |Z_s|^2 + \varphi_s(s, \bar{Y}_s) \right] ds \\ &\quad + \varphi_x(s, \bar{Y}_s) \bar{Z}_s dB_s, \quad s \in [0, T]. \end{aligned}$$

Then, in view of $|Y_s| \leq \bar{Y}_s$, by Proposition 3.3 we have

$$d\varphi(s, \bar{Y}_s) \geq \varphi_x(s, \bar{Y}_s) \bar{Z}_s dB_s, \quad s \in [0, T]. \quad (3.22)$$

For each $n \geq 1$ and $t \in [0, T]$, define the following stopping time

$$\tau_n^t := \inf \left\{ s \in [t, T] : \int_t^s [\varphi_x(r, \bar{Y}_r)]^2 |\bar{Z}_r|^2 dr \geq n \right\} \wedge T$$

with the convention that $\inf \emptyset = +\infty$. It follows from the definition of τ_n^t and (3.22) that for each $n \geq 1$ and $t \in [0, T]$,

$$\varphi(t, \bar{Y}_t) \leq \mathbb{E} [\varphi(\tau_n^t, \bar{Y}_{\tau_n^t}) | \mathcal{F}_t]. \quad (3.23)$$

On the other hand, observe from (3.18) and (3.19) that

$$\frac{1}{2}(k_0 + x) \leq \varphi(s, x) \leq k_1(k_0 + x), \quad (s, x) \in [0, T] \times [0, +\infty), \quad (3.24)$$

where k_0 is defined in (3.18) and

$$k_1 := \exp \left[2 \left(\beta + \frac{4\gamma^2}{2\lambda - 1} \right) T \right]. \quad (3.25)$$

Combining (3.23) and (3.24) yields that for each $n \geq 1$ and $t \in [0, T]$,

$$\frac{1}{2}(k_0 + \bar{Y}_t) \leq \varphi(t, \bar{Y}_t) \leq \mathbb{E} [\varphi(\tau_n^t, \bar{Y}_{\tau_n^t}) | \mathcal{F}_t] \leq k_1 \mathbb{E} [(k_0 + \bar{Y}_{\tau_n^t}) | \mathcal{F}_t].$$

Since $\bar{Y}$ belongs to class (D) and $\tau_n^t \rightarrow T$ as $n \rightarrow \infty$ for each $t \in [0, T]$, by sending n to infinity in the last inequality we deduce that

$$\bar{Y}_t \leq 2k_1 k_0 + 2k_1 \mathbb{E} [\bar{Y}_T | \mathcal{F}_t] - k_0, \quad t \in [0, T],$$

which is the desired assertion. $\square$

Before proving Theorem 2.1, we introduce the following proposition, which is a direct corollary of Proposition 3.8 in [14].

Proposition 3.5. Assume that the generator g satisfies (H2) and $(Y_t, Z_t)_{t \in [0, T]}$ is a solution of BSDE(ξ, g). If $|Y_t| + \int_0^t f_s ds \in \mathcal{S}^p$ for $p > 0$, then $Z \in \mathcal{M}^q$ for each $q \in (0, p)$.

Proof of Theorem 2.1. For positive integers $n, p \geq 1$, let

$$\xi^{n,p} := \xi^+ \wedge n - \xi^- \wedge p \quad \text{and}$$

$$g^{n,p}(\omega, t, y, z) := g^+(\omega, t, y, z) \wedge n - g^-(\omega, t, y, z) \wedge p.$$

As the terminal condition $\xi^{n,p}$ is bounded and $g^{n,p}$ is a bounded generator, it follows from [7] that BSDE($\xi^{n,p}, g^{n,p}$) admits a maximal bounded solution $(Y^{n,p}, Z^{n,p})$ such that $Y^{n,p}$ is bounded and $Z^{n,p} \in \mathcal{M}^2$. It is not hard to verify that $g^{n,p}$ satisfies (H2) with $f \wedge (n \vee p)$ instead of f . It then follows from Proposition 3.4 that there exists a constant $C > 0$ depending only on $(\beta, \gamma, \lambda, T)$ such that for each $t \in [0, T]$ and $n, p \geq 1$,

$$\begin{aligned} |Y_t^{n,p}| &\leq |Y_t^{n,p}| + \int_0^t [f_s \wedge (n \vee p)] dt \\ &\leq C \mathbb{E} \left[|\xi^{n,p}| + \int_0^T [f_t \wedge (n \vee p)] dt \middle| \mathcal{F}_t \right] + C \\ &\leq C \mathbb{E} \left[|\xi| + \int_0^T f_t dt \middle| \mathcal{F}_t \right] + C. \end{aligned} \quad (3.26)$$

It follows from Theorem 2.3 in [2] that $Y^{n,p}$ is nondecreasing in n and non-increasing in p . Then, in view of (3.26) and assumptions (H1) and (H3), by using the localization technique put forward initially in [9] we deduce that there exists an $(\mathcal{F}_t)$ -adapted process $(Z_t)_{t \in [0, T]}$ such that $(Y, Z) := (\inf_p \sup_n Y^{n,p}, Z)$ is a solution to BSDE(ξ, g). Furthermore, by letting n and p to infinity in (3.26) yields the desired inequality (2.1), and then the process Y belongs to class (D).

Let us further show that $(Y, Z) \in \mathcal{S}^p \times \mathcal{M}^p$ for each $p \in (0, 1)$. Indeed, in view of (2.1), it follows from Lemma 6.1 in [1] that for each $p \in (0, 1)$,

$$\begin{aligned} &\mathbb{E} \left[\sup_{t \in [0, T]} \left(|Y_t| + \int_0^t f_s ds \right)^p \right] \\ &\leq (2C)^p \left[\frac{1}{1-p} \left(\mathbb{E} \left[|\xi| + \int_0^T f_s ds \right] \right)^p + 1 \right] < +\infty, \end{aligned}$$

which means that $Y \in \mathcal{S}^p$ for each $p \in (0, 1)$, and then by Proposition 3.5, $Z \in \mathcal{M}^p$ for each $p \in (0, 1)$. Thus, the existence part of Theorem 2.1 is proved.

The uniqueness part is a direct corollary of Proposition 2.5, which will be proved below. The proof of Theorem 2.1 is then complete. $\square$

Proof of Proposition 2.5. We only prove the case that the generator g satisfies (H4)–(H5), and $d\mathbb{P} \times dt - a.e.$,

$$\mathbf{1}_{Y_t > Y'_t} (g(t, Y'_t, Z'_t) - g(t, Y_t, Z_t)) \leq 0. \quad (3.27)$$

The other case can be proved in the same way. Without loss of generality, we always assume that the function $\rho(\cdot)$ in (H4) and the function $\kappa(\cdot)$ in (H5) verify that for each $x \geq 0$,

$$\rho(x) \leq Ax + A \quad \text{and} \quad \kappa(x) \leq Ax + A, \quad (3.28)$$

where $A > 0$ is a universal constant.

Define $\hat{Y} := Y - Y'$ and $\hat{Z} := Z - Z'$. Then, $(\hat{Y}, \hat{Z})$ verifies

$$\begin{aligned} \hat{Y}_t &= \xi - \xi' + \int_t^T (g(s, Y_s, Z_s) - g(s, Y'_s, Z'_s)) ds \\ &\quad - \int_t^T \hat{Z}_s dB_s, \quad t \in [0, T]. \end{aligned} \quad (3.29)$$

Let

$$\tilde{Y}_t := \hat{Y}_t^+ + 2At \quad \text{and} \quad \tilde{Z}_t := \mathbf{1}_{\hat{Y}_t > 0} \hat{Z}_t, \quad t \in [0, T].$$

Itô–Tanaka's formula yields

$$\begin{aligned}\tilde{Y}_t = \tilde{Y}_T + \int_t^T (\mathbf{1}_{\tilde{Y}_s > 0} (g(s, Y_s, Z_s) - g(s, Y'_s, Z'_s)) - 2A) ds \\ - \int_t^T \tilde{Z}_s dB_s - \int_t^T dL_s, \quad t \in [0, T],\end{aligned}$$

where L is the local time of $\hat{Y}$ at 0.

Furthermore, in view of assumptions (H4) and (H5) of the generator g together with (3.27) and (3.28), we have $d\mathbb{P} \times ds - a.e.$,

$$\begin{aligned}\mathbf{1}_{\tilde{Y}_s > 0} (g(s, Y_s, Z_s) - g'(s, Y'_s, Z'_s)) - 2A \\ \leq \rho(\hat{Y}_s^+) + \mathbf{1}_{\tilde{Y}_s > 0} \kappa \left(\frac{|\tilde{Z}_s|}{(\ln(e + |\tilde{Z}_s|))^\lambda} \right) - 2A \\ \leq A\hat{Y}_s^+ + A \frac{\mathbf{1}_{\tilde{Y}_s > 0} |\tilde{Z}_s|}{(\ln(e + |\tilde{Z}_s|))^\lambda} \leq A\tilde{Y}_s + \frac{A|\tilde{Z}_s|}{(\ln(e + |\tilde{Z}_s|))^\lambda}.\end{aligned}\quad (3.30)$$

Let the function $\varphi(\cdot, \cdot)$ be defined in Proposition 3.3 with A instead of β and γ in (3.18) and (3.19). In view of (3.30), applying Itô–Tanaka's formula to the process $\varphi(s, \tilde{Y}_s)$ yields that

$$\begin{aligned}d\varphi(s, \tilde{Y}_s) = -\varphi_x(s, \tilde{Y}_s) (\mathbf{1}_{\tilde{Y}_s > 0} (g(s, Y_s, Z_s) - g(s, Y'_s, Z'_s)) - 2A) ds \\ + \varphi_x(s, \tilde{Y}_s) \tilde{Z}_s dB_s \\ + \varphi_x(s, \tilde{Y}_s) dL_s + \frac{1}{2} \varphi_{xx}(s, \tilde{Y}_s) |\tilde{Z}_s|^2 ds + \varphi_s(s, \tilde{Y}_s) ds \\ \geq \left[-\varphi_x(s, \tilde{Y}_s) \left(A\tilde{Y}_s + \frac{A|\tilde{Z}_s|}{(\ln(e + |\tilde{Z}_s|))^\lambda} \right) \right. \\ \left. + \frac{1}{2} \varphi_{xx}(s, \tilde{Y}_s) |\tilde{Z}_s|^2 + \varphi_s(s, \tilde{Y}_s) \right] ds \\ + \varphi_x(s, \tilde{Y}_s) \tilde{Z}_s dB_s, \quad s \in [0, T].\end{aligned}$$

Then, by (3.20) in Proposition 3.3 with A instead of β and γ we have

$$d\varphi(s, \tilde{Y}_s) \geq \varphi_x(s, \tilde{Y}_s) \tilde{Z}_s dB_s, \quad s \in [0, T].$$

With the last inequality in hand, noticing that $\tilde{Y}$ belongs to class (D) and using a same argument as that in the proof of Proposition 3.4 we deduce that for each $t \in [0, T]$,

$$\tilde{Y}_t \leq 2k_1 k_0 + 2k_1 \mathbb{E}[\tilde{Y}_T | \mathcal{F}_t] - k_0,$$

where the constants k_0 and k_1 are defined respectively in (3.18) and (3.25) with A instead of β and γ . Then, we have for each $t \in [0, T]$,

$$\begin{aligned}(Y_t - Y'_t)^+ + 2At \leq 2k_1 k_0 + 2k_1 (\mathbb{E}[(\xi - \xi')^+ | \mathcal{F}_t] + 2AT) - k_0 \\ \leq 2k_1 k_0 + 4Ak_1 T,\end{aligned}$$

which means that $(Y - Y')^+$ is a bounded process.

Finally, in view of (H4)–(H5) of g , $\xi \leq \xi'$ and (3.27) together with the fact that $(Y - Y')^+$ is a bounded process, we can apply Theorem 2.1 in [2] to obtain the desired assertion. $\square$

CRediT authorship contribution statement

Shengjun Fan: Conceptualization, Methodology, Investigation.
Ying Hu: Conceptualization, Methodology, Investigation. **Shan-jian Tang:** Conceptualization, Methodology, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

Acknowledgments

The authors would like to thank the anonymous referees for their careful reading and valuable suggestions.

References

- [1] P. Briand, B. Delyon, Y. Hu, E. Pardoux, L. Stoica, L^p solutions of backward stochastic differential equations, *Stochastic Process. Appl.* 108 (1) (2003) 109–129.
- [2] S. Fan, Bounded solutions, L^p ($p > 1$) solutions and L^1 solutions for one-dimensional BSDEs under general assumptions, *Stochastic Process. Appl.* 126 (2016) 1511–1552.
- [3] E. Pardoux, S. Peng, Adapted solution of a backward stochastic differential equation, *Syst. Control Lett.* 14 (1) (1990) 55–61.
- [4] N. El Karoui, S. Peng, M.C. Quenez, Backward stochastic differential equations in finance, *Math. Finance* 7 (1) (1997) 1–71.
- [5] J.-P. Lepeltier, J. San Martin, Backward stochastic differential equations with continuous coefficient, *Statist. Probab. Lett.* 32 (4) (1997) 425–430.
- [6] S. Fan, L. Jiang, L^p ($p > 1$) Solutions for one-dimensional BSDEs with linear-growth generators, *J. Appl. Math. Comput.* 38 (1–2) (2012) 295–304.
- [7] M. Kobylanski, Backward stochastic differential equations and partial differential equations with quadratic growth, *Ann. Probab.* 28 (2) (2000) 558–602.
- [8] F. Delbaen, Y. Hu, X. Bao, Backward SDEs with superquadratic growth, *Probab. Theory Related Fields* 150 (24) (2011) 145–192.
- [9] P. Briand, Y. Hu, BSDE with quadratic growth and unbounded terminal value, *Probab. Theory Related Fields* 136 (4) (2006) 604–618.
- [10] Y. Hu, S. Tang, Existence of solution to scalar BSDEs with $L \exp \sqrt{\frac{2}{\lambda}} \log(1+L)$ -integrable terminal values, *Electron. Commun. Probab.* 23 (2018) Paper No. 27, 11pp.
- [11] S. Fan, Y. Hu, Existence and uniqueness of solution to scalar BSDEs with $L \exp(\mu \sqrt{2 \log(1+L)})$ -integrable terminal values: the critical case, *Electron. Commun. Probab.* 24 (2019) Paper No. 49, 10pp.
- [12] S. Tang, Dual representation as stochastic differential games of backward stochastic differential equations and dynamic evaluations, *C. R. Math. Acad. Sci. Paris* 342 (2006) 773–778.
- [13] S. Fan, Y. Hu, Well-posedness of scalar BSDEs with sub-quadratic generators and related PDEs, *Stochastic Process. Appl.* 131 (2021) 21–50.
- [14] S. Fan, Y. Hu, S. Tang, Existence, uniqueness and comparison theorem on unbounded solutions of scalar super-linear BSDEs, *Stochastic Process. Appl.* 157 (2023) 335–375.
- [15] R. Buckdahn, Y. Hu, S. Tang, Uniqueness of solution to scalar BSDEs with $L \exp(\mu \sqrt{2 \log(1+L)})$ -integrable terminal values, *Electron. Commun. Probab.* 23 (2018) Paper No. 59, 8pp.
- [16] O. H. M.-C. Kim, C.-G. Pak, Uniqueness of solution to scalar BSDEs with $L \exp(\mu_0 \sqrt{2 \log(1+L)})$ -integrable terminal values: an L^1 -solution approach, *C. R. Math. Acad. Sci. Paris* 359 (9) (2021) 1085–1095.
- [17] P. Briand, Y. Hu, Quadratic BSDEs with convex generators and unbounded terminal conditions, *Probab. Theory Related Fields* 141 (3) (2008) 543–567.
- [18] F. Delbaen, Y. Hu, A. Richou, On the uniqueness of solutions to quadratic BSDEs with convex generators and unbounded terminal conditions, *Ann. Inst. Henri Poincaré Probab. Stat.* 47 (2011) 559–574.
- [19] S. Fan, Y. Hu, S. Tang, On the uniqueness of solutions to quadratic BSDEs with non-convex generators and unbounded terminal conditions, *C. R. Math. Acad. Sci. Paris* 358 (2) (2020) 227–235.
- [20] S. Peng, Backward SDE and related g -expectation, backward stochastic differential equations, in: N. El Karoui, L. Mazliak (Eds.), *Pitman Res. Notes Math. Ser. Vol. 364*, Longman, Harlow, 1997, pp. 141–159.
- [21] S. Peng, Monotonic limit theorem of BSDE and nonlinear decomposition theorem of Doob–Meyer's type, *Probab. Theory Related Fields* 113 (4) (1999) 23–30.
- [22] L. Jiang, Convexity, translation invariance and subadditivity for g -expectations and related risk measures, *Ann. Appl. Probab.* 18 (1) (2008) 245–258.